

Magentus Coding Challenge: Toy Robot Simulator

Description:

The task is to create a simulation of a toy robot moving on a square tabletop of dimensions 5 units x 5 units.

- There are no obstructions on the table surface.
- The robot is free to move around the surface, but must not fall off the table.
- Any movement that would cause the robot to fall must be ignored, while valid commands should still be accepted and executed.

Application Requirements

Your application must be able to read and process the following commands:

- PLACE X,Y,F
- MOVE
- LEFT
- RIGHT
- REPORT

Command Details

- PLACE X,Y,F
 - Places the robot on the table at coordinates (X, Y), facing direction F (NORTH, SOUTH, EAST, or WEST).
 - The origin (0,0) represents the south-west corner of the table.
 - The first valid command must be a PLACE command.
 - Any commands before a valid PLACE should be ignored.
 - Additional PLACE commands can be used to reposition the robot.
- MOVE
 - Moves the robot one unit forward in the direction it is currently facing.
- LEFT / RIGHT
 - Rotates the robot 90° in the specified direction without changing its position.
- REPORT
 - Outputs the current X, Y, and F (facing direction) of the robot.
 - Output format is flexible, but standard console output is sufficient.

Constraints

- The robot must not fall off the table.
- Any move that would result in the robot falling should be ignored.
- A robot not yet placed on the table should ignore all commands except PLACE.
- Input may be provided via file or standard input, at your discretion.
- Please include test data to demonstrate your solution.

Example Input and Output

Example A

- PLACE 0,0,NORTH
- MOVE
- REPORT
- Output: 0,1,NORTH

Example B

- PLACE 0,0,NORTH
- LEFT
- REPORT
- Output: 0,0,WEST

Example C

- PLACE 1,2,EAST
- MOVE
- MOVE
- LEFT
- MOVE
- REPORT
- Output: 3,3,NORTH